

07

Wednesday

Homework - 5

April

Day (097 - 268)

08.00

1) Given

09.00

$$y = X\beta + \sigma \epsilon$$

$$\epsilon \sim N(0, I_n)$$

10.00

$$n = 200$$

$$p = 5$$

11.00

To show, if $v = P\epsilon \Rightarrow$ where P is orthogonal
 $\epsilon \sim N(0, I_n)$

12.00

then $E(v_i) = 0$

$$\text{Var}(v_i) = 1 \text{ for all } i = 1, \dots, n$$

13.00

$$v = P\epsilon \Rightarrow E(v) = E(P\epsilon)$$

14.00

Using linearity, $\Rightarrow P E(\epsilon)$

15.00

As $E(\epsilon) = 0 \Rightarrow E(v_i) = 0 //$

16.00

17.00

$$\text{Var}(v) = \text{Var}(P\epsilon)$$

$$= P \text{Var}(\epsilon) P^T$$

18.00

Since $\text{Var}(\epsilon) = I$

Notes

$$\Rightarrow \text{Var}(v) = P I P^T$$

$$= P P^T$$

2021	March	2021	April
S M T W T F S	S M T W T F S	S M T W T F S	
1 2 3 4 5 6		1 2 3	
7 8 9 10 11 12 13		4 5 6 7 8 9 10	
14 15 16 17 18 19 20		11 12 13 14 15 16 17	
21 22 23 24 25 26 27		18 19 20 21 22 23 24	
28 29 30 31		25 26 27 28 29 30	

08.00

Since P is orthogonal,

09.00

$$PP^T = I$$

$$\Rightarrow \text{Var}(y) = I$$

10.00

$$\Rightarrow \text{Var}(y_i) = 1 //$$

11.00

12.00

$$\Rightarrow E(y_i) = 0, \text{Var}(y_i) = 1$$

for all $i = 1, \dots, n$

13.00

14.00

15.00

16.00

17.00

18.00

Notes

e

2) To show, if $y = Q\beta$,

Then $PX\beta = Dy$

We know from SVD:

$$PXQ^T = D \Rightarrow PXQ^TQ = DQ$$

Multiply both sides by β , $\Rightarrow PXQ^TQ\beta = DQ\beta$

We know that $y = Q\beta$ & $Q^TQ = I$

$$\Rightarrow PX\beta = Dy //$$

2021	May	2021	June
S M T W T F S	S M T W T F S	S M T W T F S	
30 31	1	1 2 3 4 5	
2 3 4 5 6 7 8	6 7 8 9 10 11 12		
9 10 11 12 13 14 15	13 14 15 16 17 18 19		
16 17 18 19 20 21 22	20 21 22 23 24 25 26		
23 24 25 26 27 28 29	27 28 29 30		

09

Friday

Day (099 - 266)

April 2

08.00

For transforming the model,

09.00

$$y = X\beta + \sigma \epsilon$$

10.00

Multiply by $P \Rightarrow Py = PX\beta + \sigma P\epsilon$

11.00

We know, $z = Py$ & $v = P\epsilon$

12.00

$$\Rightarrow z = D\gamma + \sigma v$$

13.00

Since $D = \begin{pmatrix} \Lambda \\ 0 \end{pmatrix}$, we get,

14.00

$$z_i = \begin{cases} \lambda_i \gamma_i + \sigma v_i, & i=1, \dots, p \\ \sigma v_i, & i=p+1, \dots, n \end{cases}$$

15.00

16.00

17.00

1) 3) To prove,

18.00

$$\sum_{j=1}^p \text{Var}(\hat{\beta}_j) = \sigma^2 \sum_{j=1}^p \frac{1}{\lambda_j^2}$$

Notes

2021	March	2021	April
S M T W T F S		S M T W T F S	
1 2 3 4 5 6		1 2 3	
7 8 9 10 11 12 13		4 5 6 7 8 9 10	
14 15 16 17 18 19 20		11 12 13 14 15 16 17	
21 22 23 24 25 26 27		18 19 20 21 22 23 24	
28 29 30 31		25 26 27 28 29 30	

2021 April

Week - 15

Day (100 - 265)

10

Saturday

We know

$$\text{Var}(\hat{\beta}) = \sigma^2 (X^T X)^{-1}$$

Using Eigen Decomposition,

$$X^T X = Q^T \Lambda^2 Q$$

$$\text{Then } (X^T X)^{-1} = Q^T \Lambda^{-2} Q$$

Sum of Variances = trace,

$$\Rightarrow \sum \text{Var}(\hat{\beta}_i) = \sigma^2 \text{tr}((X^T X)^{-1})$$

$$= \sigma^2 \text{tr}(Q^T \Lambda^{-2} Q)$$

Trace invariant under similarity,

$$= \sigma^2 \text{tr}(\Lambda^{-2}) \quad (\text{Since } Q^T Q = I)$$

$$\Rightarrow \sum_{j=1}^p \text{Var}(\hat{\beta}_i) = \sigma^2 \sum_{j=1}^p \frac{1}{\lambda_j^2}$$

Week 16

11

Sunday

Notes

X

2021	May	2021	June
S M T W T F S	S M T W T F S	S M T W T F S	
30 31	1	1 2 3 4 5	
2 3 4 5 6 7 8	6 7 8 9 10 11 12		
9 10 11 12 13 14 15	13 14 15 16 17 18 19		
16 17 18 19 20 21 22	20 21 22 23 24 25 26		
23 24 25 26 27 28 29	27 28 29 30		

12

Monday

Day (102 - 263)

April 2021

08.00

1) 4) From the dataset provided, compute $\hat{y}$

09.00

We know that $y = Q\beta$

10.00

$$\Rightarrow \hat{y} = Q\hat{\beta}$$

11.00

We know, $\hat{\beta} = (X^T X)^{-1} X^T y$ (from OLS estimator)

12.00

$$\Rightarrow \hat{y} = Q(X^T X)^{-1} X^T y$$

13.00

We also know, $(X^T X)^{-1} = Q^T \Lambda^{-2} Q$

14.00

$$\Rightarrow \hat{y} = Q(Q^T \Lambda^{-2} Q) X^T y$$

15.00

$$\Rightarrow \hat{y} = \Lambda^{-2} Q X^T y \quad (\text{since } QQ^T = I)$$

16.00

where Q = eigenvector matrix of $X^T X$

17.00

Λ^2 = eigenvalues of $X^T X$

$\Lambda^{-2} = \text{diag}(1/\lambda_1^2, \dots, 1/\lambda_p^2)$

18.00

Notes

2021	March	2021	April
S M T W T F S		S M T W T F S	
1 2 3 4 5 6		1 2 3	
7 8 9 10 11 12 13		4 5 6 7 8 9 10	
14 15 16 17 18 19 20		11 12 13 14 15 16 17	
21 22 23 24 25 26 27		18 19 20 21 22 23 24	
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Now, computing this in R code.

```
data <- read.csv("HW5_data.csv")
```

```
X <- as.matrix(data[, -ncol(data)])
```

```
y <- as.matrix(data[, ncol(data)])
```

```
n <- nrow(X)
```

```
p <- ncol(X)
```

Step 1: Compute beta-hat (OLS estimator)

```
beta-hat <- solve(t(X) %*% X) %*% t(X) %*% y
```

Step 2: Eigen decomposition of $X^T X$

```
eig <- eigen(t(X) %*% X)
```

Extract eigenvalues & eigenvectors

```
lambda-sq <- eig$values
```

```
Q <- eig$vectors
```

Step 3: Compute gamma-hat

```
gamma-hat <- Q %*% beta-hat
```

$$\Rightarrow \hat{\gamma} = \begin{bmatrix} -0.103 \\ -0.341 \\ -0.398 \\ 0.295 \\ 0.189 \end{bmatrix}$$

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9 10 11 12 13 14 15	13 14 15 16 17 18 19		
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23 24 25 26 27 28 29	27 28 29 30		

14

Day (104 - 261)

Wednesday

08.00

5) We need smallest m such that,

09.00

$$\frac{\sum_{j=1}^m \lambda_j^2}{\sum_{j=1}^p \lambda_j^2} > 0.9$$

10.00

11.00

and the corresponding $\hat{\beta}_{PCA}$

12.00

We know $\hat{\beta}_{PCA} = Q_m \Lambda_m^{-1} y_m$

13.00

Rcode:

14.00

Code for loading data & storing values in X, y, n, p
variables are the same

15.00

16.00

Step1: Eigen decomposition of $X^T X$
 $eig \leftarrow \text{eigen}(t(X) \%*\% X)$

17.00

$\text{lambda_sq} \leftarrow eig\$values$
 $Q \leftarrow eig\$vectors$

18.00

Notes

Sort in descending order
 $idx \leftarrow \text{order}(\text{lambda_sq}, \text{decreasing} = \text{TRUE})$
 $\text{lambda_sq} \leftarrow \text{lambda_sq}[idx]$
 $Q \leftarrow Q[, idx]$

2021

March

2021

April

S	M	T	W	T	F	S	S	M	T	W	T	F	S
1	2	3	4	5	6				1	2	3		
7	8	9	10	11	12	13	4	5	6	7	8	9	10
14	15	16	17	18	19	20	11	12	13	14	15	16	17
21	22	23	24	25	26	27	18	19	20	21	22	23	24
28	29	30	31				25	26	27	28	29	30	

2021 April

Week - 16

Day (105 - 260)

15

Thursday

Step 2: Find smallest m

$$\text{cum_ratio} \leftarrow \text{cumsum}(\text{lambda_sq}) / \text{sum}(\text{lambda_sq})$$

$$m \leftarrow \text{which}(\text{cum_ratio} > 0.9)[1]$$

$$m = 3$$

Step 3: Compute OLS beta-hat

$$\text{beta_hat} \leftarrow \text{solve}(t(X) \%*\% X) \%*\% t(X) \%*\% y$$

Step 4: Compute Gamma-hat

$$\text{gamma_hat} \leftarrow Q \%*\% \text{beta_hat}$$

Step 5: keep only first m components

$$Q_m \leftarrow Q[, 1:m]$$

$$\text{gamma_m} \leftarrow \text{gamma_hat}[1:m]$$

Step 6: Compute PCA beta

$$\text{beta_pca} \leftarrow Q_m \%*\% \text{gamma_m}$$

$$\Rightarrow \text{beta_pca} = \begin{bmatrix} -0.239 \\ 0.177 \\ 0.259 \\ 0.314 \\ 0.176 \end{bmatrix}$$

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30 31	1	1 2 3 4 5	
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9 10 11 12 13 14 15	13 14 15 16 17 18 19		
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23 24 25 26 27 28 29	27 28 29 30		

16

Friday

April 2021

Week - 16

Day (106 - 259)

08.00

Prior

2) Given, $\pi(\beta) = 0.5\delta_0 + 0.5N(0, \tau^2)$

09.00

Bayesian linear model,

10.00

$$y_i \sim N(x_i\beta, \sigma^2), \quad i=1, \dots, n$$

11.00

where $\sum_{i=1}^n x_i = 0$, $\sum_{i=1}^n x_i^2 = n$ & $\sum_{i=1}^n x_i y_i = \gamma$

12.00

13.00

Posterior density for β ,

14.00

$$\pi(\beta|y) \propto \left(\prod_{i=1}^n N(y_i|x_i\beta, \sigma^2) \right) [0.5\delta_0 + 0.5N(\beta|0, \tau^2)]$$

15.00

16.00

Since, Always Posterior = Likelihood x Prior

17.00

How, Likelihood can also be written as,

18.00

$$L(\beta) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(y_i - x_i\beta)^2}{2\sigma^2}\right)$$

Notes

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2021 April

Week - 16

Day (107 - 258)

17

Saturday

2) We must compute,

Marginal Likelihood,

$$m(y) = \int \left(\prod_{i=1}^n N(y_i | x_i \beta, \sigma^2) \right) N(\beta | 0, \tau^2) d\beta$$

We know

$$\begin{aligned} \sum (y_i - x_i \beta)^2 &= \sum y_i^2 - 2\beta \sum x_i y_i + \beta^2 \sum x_i^2 \\ &= \sum y_i^2 - 2\beta y + n\beta^2 \end{aligned}$$

So exponent becomes, $\exp\left(-\frac{1}{2\sigma^2} [\sum y_i^2 - 2\beta y + n\beta^2]\right)$

Multiply by prior density, $\exp\left(-\frac{\beta^2}{2\tau^2}\right)$

Combine quadratic terms: $\beta^2 \left(\frac{n}{\sigma^2} + \frac{1}{\tau^2}\right) - \frac{2y}{\sigma^2} \beta$

We also know, $A = \frac{n}{\sigma^2} + \frac{1}{\tau^2}$

Posterior variance $\psi^2 = \frac{1}{A} = \left(\frac{n}{\sigma^2} + \frac{1}{\tau^2}\right)^{-1}$

Posterior mean, $\nu = \frac{\psi^2 y}{\sigma^2}$

Week 17 18
Sunday

2

2021	May	2021	June
S M T W T F S		S M T W T F S	
30 31	1	1 2 3 4 5	
2 3 4 5 6 7 8	6 7 8 9 10 11 12		
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16 17 18 19 20 21 22	20 21 22 23 24 25 26		
23 24 25 26 27 28 29	27 28 29 30		

Monday

08.00 After completing the square, the integral becomes,

09.00
$$m(y) = \left(\prod N(y_i | 0, \sigma^2) \right) \sqrt{\frac{\psi^2}{\tau^2}} \exp\left(\frac{y^2}{2\sigma^2(n + \sigma^2/\tau^2)}\right)$$

10.00

11.00

12.00

13.00 3) Now plug marginal likelihood into posterior:

14.00
$$\pi(\beta | y) \propto \left(\prod N(y_i | 0, \sigma^2) \right) S_0 + m(y) N(\beta | y, \psi^2)$$

15.00

where,
$$\psi^2 = \left(\frac{n}{\sigma^2} + \frac{1}{\tau^2} \right)^{-1}$$

16.00

17.00
$$V = \frac{\psi^2 y}{\sigma^2}$$

18.00

(The derivation is shown in the previous questions answer)

Notes

2021	March	2021	April
S M T W T F S	S M T W T F S	S M T W T F S	
1 2 3 4 5 6	1 2 3		
7 8 9 10 11 12 13	4 5 6 7 8 9 10		
14 15 16 17 18 19 20	11 12 13 14 15 16 17		
21 22 23 24 25 26 27	18 19 20 21 22 23 24		
28 29 30 31	25 26 27 28 29 30		

2021 April

Week - 17

Day (110 - 255)

20

Tuesday

4) Posterior Probability $\beta=0$.

08.00

$$\Rightarrow P(\beta=0|y) = \frac{\prod N(y_i|0, \sigma^2)}{\prod N(y_i|0, \sigma^2) + m(y)}$$

10.00

We can substitute $m(y)$ from previous solutions

12.00

13.00

5) Significance of Ratio $\frac{m(y)}{\prod N(y_i|0, \sigma^2)}$

14.00

15.00

$\Rightarrow$ Bayes factor = Evidence under M_1 / Evidence under M_0

16.00

This is the Bayes factor comparing,

$\rightarrow$ Model 1: $\beta \neq 0$ (Model with regression effect)

$\rightarrow$ Model 0: $\beta = 0$ (Model with no regression effect)

17.00

As,

$$y^2 \rightarrow \infty$$

18.00

the exponential term dominates

$$\exp(y^2/\text{constant}) \rightarrow \infty$$

Notes

$\Rightarrow$ Bayes factor $\rightarrow \infty$

Meaning ~~over~~ overwhelming evidence that $\beta \neq 0$.

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30 31	1	1 2 3 4 5	
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9 10 11 12 13 14 15	13 14 15 16 17 18 19		
16 17 18 19 20 21 22	20 21 22 23 24 25 26		
23 24 25 26 27 28 29	27 28 29 30		